

Sachith Gunasekara

Curriculum Vitae · AI Research and Engineering · Colombo, Sri Lanka

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EDUCATION

BSc (Hons) in Computer Science

02/2019 – 07/2023

University of Kelaniya · Dalugama, Sri Lanka

- First Class. Ranked 1 of 125. CGPA 3.78/4.00; Computer Science subject GPA 3.86/4.00.
- Admitted to the Computer Science honours stream on merit (subject GPA 3.87). Minors in Statistics and Pure Mathematics (years 1–2).
- Thesis: *Engraving sentence-level positional data into N-gram profiles for document classification*

Diploma in Software Engineering

10/2017 – 04/2018

ESOFT Metro Campus · Matale, Sri Lanka

PUBLICATIONS

Pre-prints

1. **Gunasekara S.**, Ratnayake Y. (2025). Effects of structure on reasoning in instance-level Self-Discover. arXiv:2507.03347. Under revision for resubmission.
2. Dissanayake C., Lowe L., **Gunasekara S.**, Ratnayake Y. (2024). OpenBezoar: small, cost-effective and open models trained on mixes of instruction data. arXiv:2404.12195.

Conference abstracts

3. **Gunasekara L. M. S.**, Premadasa H. K. S. (2023). Capturing sentence-level positional data into N-gram profiles for document classification. *Proceedings of ICAPS 2023*, Vol. 3, p. 119, Faculty of Science, University of Kelaniya. Oral presentation; Best Presenter award.

RESEARCH & INDUSTRY EXPERIENCE

Senior AI Engineer

02/2026 – Present

Gapstars, placed at Envision AI Labs (Leiden, Netherlands) · Colombo, Sri Lanka · Hybrid

- Chose the retrieval embedding model on evidence. I built a synthetic benchmark of question–answer pairs linked to source records and compared four leading models on recall, nDCG and MRR. Significance tests confirmed the differences were real before we committed.
- Building a ground-truth benchmark from client-provided documents, so retrieval and answer quality are measured on real data instead of synthetic proxies.
- Designed the evaluation framework for RAG and agentic systems, using LLM-as-a-judge metrics, claim-level faithfulness checks and expert-validated gold-standard datasets. This gives the team a repeatable quality check before release.
- Built and benchmarked LLM-based insight-generation features on client-approved pharmaceutical and clinical data.
- Contributed to the multi-agent architecture behind narrative trend generation: hybrid retrieval, guardrails, citation grounding and context management. Every generated insight can be traced to its source documents.

Senior AI Engineer

06/2025 – 02/2026

Arcadea Group · Toronto, Canada · Remote

- Automated structured data extraction from a US state police agency's records with an offline multi-agent system. Its 10+ planning, generation, execution and evaluation agents write named-group regex patterns for field–value pairs.
 - Made accuracy verifiable with a generate–test–revise loop. Patterns grow one batch of fields at a time, run on the real documents and are scored against ground truth. Failures are revised before the next batch starts.
 - Delivered a validated, reusable regex instead of an ongoing model dependency, so the agency can extract data from future documents offline without re-running the agents.
- Built a customer-care assistant that answers new queries from two sources: company knowledge as ground truth, and historical phone calls as learned experience.
 - Compared traditional RAG with graph-based RAG to improve retrieval quality.
 - Applied topic modelling to keep a curated topic list across the whole knowledge base.
 - Built a proof of concept for verifiable traceability that returns the source records behind each generated answer.

Surge Global

03/2023 – 07/2026

Colombo, Sri Lanka

Machine Learning Engineer (full-time, then part-time contract)

04/2025 – 07/2026

- Led AI delivery of an orchestrated multi-agent system for an insurance warranty-claims intermediary.
 - Agents gather claim details from customers in natural language, check them against a RAG knowledge base and keep the conversation on track.
 - Service technicians get a ready recommendation for each claim: an issue summary, the troubleshooting steps the customer already tried and the results they saw.
- Researched connecting a real-time GPT-4o voice agent to the app's multi-agent workflows for a founder relationship assistant. The goal was letting users act in the app by voice, pulling meeting briefs, conversation history and summaries for their top 100 contacts.

Associate Machine Learning Engineer; Data Science Intern

03/2023 – 04/2025

- Tested whether LLM reasoning should use structured output. As first author, I built iSelf-Discover, an instance-level adaptation of Self-Discover, and showed that unstructured plans beat structured JSON plans across BBH, T4D and MATH, by up to 18.9% relative on MATH.
- Co-developed OpenBezoar, a low-cost recipe for small open instruction-following models. The recipe filters Falcon-40B synthetic data with GPT-4, fine-tunes OpenLLaMA 3B v2 with QLoRA, then aligns it with DPO. The final model beat many 3B-scale models on LM Eval Harness and MT-Bench, and we released the models and datasets on Hugging Face.
- Wrote nanoJAXGPT, a JAX/Equinox rewrite of nanoGPT with a Hugging Face tutorial, as a teaching resource for engineers learning JAX.

TEACHING

Instructor, AI Engineering Bootcamp

09/2025 – Present

StemLink · Online

- Teach an applied LLM curriculum that takes learners from fundamentals to working systems: transformers, prompting, agents (LangChain, LangGraph, AutoGen), RAG (LlamaIndex, Pinecone) and LoRA/QLoRA fine-tuning.
- Supervise capstone builds through to working demos, including a multi-agent travel planner and a knowledge-management MVP with retriever, summariser and evaluator agents.

INVITED TALKS & WORKSHOPS

- A Step Beyond the Hype: A Journey Through Modern AI. IEEE Student Branch, Wayamba University of Sri Lanka. Online, 04/2025.
- Implementing AI Agents. AI-Driven Sri Lanka, for university undergraduates. Online, 08/2024.

HONORS, AWARDS & FELLOWSHIPS

- Youth Delegate, Global Leadership Program, Headway Institute of Strategic Alliance, Brasenose College, University of Oxford. Partially funded fellowship on global policy dialogue and strategic innovation, 08/2025.
- Gold medal for best performance in Computer Science (2017/2018 batch), 142nd General Convocation, University of Kelaniya, 03/2024.
- Best Presenter, 4th International Conference on Applied and Pure Sciences (ICAPS 2023), University of Kelaniya, 10/2023.
- Second Prize, IEEE SA P2834 Student Challenge, Valencia, Spain, 11/2022.
- Dean's List, Faculty of Science, University of Kelaniya, for academic years 2017/18, 2019/20 and 2020/21.

SERVICE & AFFILIATIONS

- Member (pending), IEEE P2834 Working Group: Standard for Secure and Trusted Learning Systems.
- Judge, Sri Lanka Artificial Intelligence Competition (SLAIC), IEEE Young Professionals Sri Lanka, 10/2024.
- Mentor, AI Bootcamp final competition, IEEE Young Professionals Sri Lanka, 12/2023.

TECHNICAL WRITING & OPEN SOURCE

- nanoJAXGPT: A pedagogical introduction to JAX/Equinox. Hugging Face blog, 10/2024. A rewrite of nanoGPT in JAX/Equinox. Code: github.com/surgeglobal/nanoJAXGPT.

SKILLS

Research areas: LLM reasoning, instruction tuning, LLM evaluation, multi-agent systems, retrieval-augmented generation

Methods: supervised fine-tuning, LoRA/QLoRA, DPO, synthetic data generation, LLM-as-a-judge, calibration

Tools: Python, JAX/Equinox, Hugging Face, LangChain, LangGraph, AutoGen, LlamaIndex, Pinecone